

Chapter 1 Test model

- **70 Multiple Choice Questions (MCQs)**
 - **30 True/False Questions**
 - Difficulty: **70% Hard, 30% Very Hard**
 - Answers are provided **after all questions**
-

Multiple Choice Questions (1–70)

1. Which of the following best defines a *database*?
 - a. A software system that manages data
 - b. A collection of unrelated files
 - c. A collection of related data
 - d. A programming language for data manipulation
-
2. The term *mini-world* refers to:
 - a. A virtual simulation of a database
 - b. A small DBMS used for testing
 - c. A part of the real world about which data is stored
 - d. A simplified version of SQL
-
3. What is the primary function of a DBMS?
 - a. To replace operating systems
 - b. To facilitate the creation and maintenance of a computerized database
 - c. To design user interfaces only
 - d. To store only textual data
-
4. Which of the following is NOT a traditional database application?
 - a. Numeric databases
 - b. Textual databases
 - c. Multimedia databases
 - d. Payroll systems
-

5. In the UNIVERSITY database example, which of the following is a valid relationship?
- a. STUDENTs teach SECTIONs
 - b. COURSEs take STUDENTs
 - c. INSTRUCTORs teach SECTIONs
 - d. DEPARTMENTs enroll in COURSEs
-

6. The self-describing nature of a database system means:
- a. The database can write its own queries
 - b. The DBMS catalog stores metadata about the database structure
 - c. Users can rename the database at will
 - d. The database runs without any software
-

7. Program-data independence allows:
- a. Programs to be written in any language
 - b. Changes in data storage without changing application programs
 - c. Automatic deletion of unused data
 - d. Direct hardware access by users
-

8. Which characteristic enables different users to see different subsets of the database?
- a. Data redundancy
 - b. Multiple views
 - c. Physical storage
 - d. Query optimization
-

9. OLTP stands for:
- a. Online Logical Transaction Protocol
 - b. Offline Transaction Processing
 - c. Online Transaction Processing
 - d. Operational Level Transaction Protocol
-

10. Who is responsible for authorizing access to the database and monitoring its use?
- a. System analyst
 - b. Application programmer
 - c. Database administrator
 - d. Tool developer

11. Which type of end-user uses “canned transactions” repeatedly?

- a. Casual
 - b. Sophisticated
 - c. Naïve or Parametric
 - d. Stand-alone
-

12. Business analysts are best described as:

- a. Workers behind the scene
 - b. Sophisticated end-users analyzing big data
 - c. Database designers
 - d. Hardware operators
-

13. Which group designs and implements the DBMS software itself?

- a. Actors on the scene
 - b. End-users
 - c. Workers behind the scene
 - d. Parametric users
-

14. Tool developers are responsible for creating software that:

- a. Replaces the DBMS
 - b. Facilitates database modeling, testing, and UI creation
 - c. Only handles backup
 - d. Writes user reports automatically
-

15. Which is a key advantage of the database approach?

- a. Increased data redundancy
 - b. Reduced application development time
 - c. Elimination of all security needs
 - d. Mandatory use of cloud storage
-

16. Enforcing integrity constraints is a function of:

- a. Operating system
- b. DBMS

- c. Web browser
 - d. Text editor
-

17. In which scenario is a DBMS *not recommended*?

- a. Multi-user airline reservation system
 - b. Simple, static application with single-user access
 - c. Hospital patient records
 - d. University enrollment system
-

18. A DBMS may be infeasible in:

- a. Large enterprises
 - b. Embedded systems with limited storage
 - c. Data warehouses
 - d. Social media platforms
-

19. Which real-time system might avoid DBMS due to overhead?

- a. Online banking
 - b. Telephone switching systems
 - c. E-commerce websites
 - d. University portals
-

20. What does metadata in a DBMS catalog typically include?

- a. User passwords
 - b. Data structures, types, and constraints
 - c. Backup schedules
 - d. Network configuration
-

21. The conceptual view of a database is presented through:

- a. Physical storage files
 - b. A data model
 - c. Binary code
 - d. SQL syntax only
-

22. Which user category includes bank tellers performing routine transactions?

- a. Casual
 - b. Sophisticated
 - c. Parametric
 - d. Stand-alone
-

23. Social media users who post and read content are classified as:

- a. Database designers
 - b. Parametric end-users
 - c. System analysts
 - d. Workers behind the scene
-

24. Which of the following is NOT a typical DBMS functionality?

- a. Defining database structure
 - b. Loading initial data
 - c. Compiling C++ programs
 - d. Concurrent transaction processing
-

25. Transactions in a database context may involve:

- a. Only reading data
 - b. Only printing reports
 - c. Reading and updating data
 - d. Installing operating systems
-

26. The recovery subsystem in a DBMS ensures:

- a. Deleted data is permanently lost
 - b. Completed transactions are permanently recorded
 - c. All users see the same view
 - d. Queries run faster
-

27. Which is a recent database application area?

- a. Payroll processing
- b. Inventory tracking
- c. Genome databases
- d. Employee time cards

28. Data abstraction in DBMS helps to:

- a. Increase storage cost
 - b. Hide storage details from users
 - c. Expose hardware to programmers
 - d. Eliminate metadata
-

29. Concurrency control guarantees:

- a. Only one user accesses the database
 - b. Each transaction is correctly executed or aborted
 - c. All queries return instant results
 - d. Data is never updated
-

30. Which is a disadvantage of using a DBMS?

- a. Reduced redundancy
 - b. High initial investment and overhead
 - c. Better security
 - d. Easier data sharing
-

31. Persistent storage for program objects is provided by:

- a. Compiler
 - b. DBMS
 - c. Text editor
 - d. Web server
-

32. Efficient query processing is supported by:

- a. Random file naming
 - b. Storage structures like indexes
 - c. Manual sorting
 - d. Avoiding metadata
-

33. Who communicates with end-users to understand their needs during database design?

- a. Tool developers
- b. Database designers

- c. Operators
 - d. Social media users
-

34. Application programmers are responsible for:
- a. Authorizing database access
 - b. Implementing specifications from system analysts
 - c. Designing the DBMS kernel
 - d. Managing hardware maintenance
-

35. The term “database system” typically includes:
- a. Only the data
 - b. DBMS software + data (+ sometimes applications)
 - c. Only the operating system
 - d. Only SQL queries
-

36. Which is NOT listed as a main characteristic of the database approach?
- a. Self-describing nature
 - b. Program-data dependence
 - c. Data abstraction
 - d. Multiple views
-

37. In the UNIVERSITY example, a STUDENT *major*s in a DEPARTMENT. This is an example of:
- a. An attribute
 - b. A relationship
 - c. A constraint
 - d. A transaction
-

38. A “casual” end-user:
- a. Uses the database daily for fixed tasks
 - b. Accesses the database occasionally when needed
 - c. Designs the database schema
 - d. Maintains personal photo databases
-

39. Stand-alone users typically:
- a. Work in large corporations
 - b. Use packaged apps with internal databases
 - c. Manage multi-user transactions
 - d. Develop DBMS software
-

40. Which of the following supports economies of scale in database use?
- a. Duplicating data across departments
 - b. Consolidating data and applications
 - c. Using separate files for each user
 - d. Avoiding metadata
-

41. A DBMS provides multiple interfaces to:
- a. Hide data from all users
 - b. Serve different classes of users
 - c. Replace programming languages
 - d. Increase hardware costs
-

42. Which function is handled by the DBMS recovery subsystem?
- a. Preventing unauthorized logins
 - b. Restoring database after failure
 - c. Drawing ER diagrams
 - d. Compiling Java code
-

43. Deductive and active rules in a DBMS can be used to:
- a. Format hard drives
 - b. Draw inferences and trigger actions from stored data
 - c. Replace human analysts
 - d. Encrypt all data automatically
-

44. Enforcing standards in large organizations includes standardizing:
- a. Only SQL syntax
 - b. Data item names, report structures, web layouts, etc.
 - c. User passwords only
 - d. Hardware vendors

45. Which is a very hard limitation that may make a DBMS insufficient?

- a. Need for simple reports
- b. Modeling limitations for complex genome data
- c. Single-user access
- d. Static data

46. GIS and location-based services may require:

- a. Standard relational DBMS only
- b. Special operations not supported by general DBMS
- c. No database at all
- d. Only textual data

47. The DBMS catalog is also known as:

- a. Data dictionary
- b. User manual
- c. Backup log
- d. Query cache

48. Which user type is “thoroughly familiar with system capabilities” and uses analytical tools?

- a. Naïve
- b. Casual
- c. Sophisticated
- d. Stand-alone

49. Mobile app users mostly fall under which category?

- a. Sophisticated
- b. Parametric
- c. Database administrators
- d. Tool developers

50. What ensures that concurrent transactions do not corrupt data?

- a. Data redundancy

- b. Concurrency control
 - c. File compression
 - d. User training
-

51. Which is NOT a responsibility of the DBA?

- a. Authorizing access
 - b. Monitoring efficiency
 - c. Writing end-user mobile apps
 - d. Acquiring software resources
-

52. The conceptual data model in the UNIVERSITY example is likely based on:

- a. Relational algebra
 - b. Entity-Relationship model
 - c. Binary trees
 - d. Machine learning
-

53. A prerequisite COURSE relationship is an example of:

- a. A unary relationship
 - b. A ternary relationship
 - c. A composite attribute
 - d. A derived key
-

54. Which of the following is a “worker behind the scene”?

- a. End-user
 - b. Database designer
 - c. System implementor of DBMS modules
 - d. Parametric clerk
-

55. What is a major part of database applications involving hundreds of transactions per second?

- a. Batch processing
- b. OLTP
- c. Data mining
- d. File archiving

56. Which advantage directly results from reduced redundancy?

- a. Higher storage cost
 - b. Inconsistent data
 - c. Lower maintenance effort
 - d. Slower queries
-

57. Persistent storage in DBMS is crucial for:

- a. Temporary variables
 - b. Program objects that must survive beyond program execution
 - c. Cache memory
 - d. Screen displays
-

58. Query optimization is provided to:

- a. Make queries longer
 - b. Ensure efficient processing
 - c. Hide data from users
 - d. Increase CPU usage
-

59. Which is a very hard scenario where DBMS overhead is unacceptable?

- a. University grade entry
 - b. Telephone switching systems
 - c. Online shopping
 - d. Social media posting
-

60. In embedded systems, a general-purpose DBMS may not fit due to:

- a. Too many users
 - b. Limited storage
 - c. High bandwidth
 - d. Excessive metadata
-

61. The ability to evolve database structure with new requirements reflects:

- a. Rigidity
- b. Flexibility

- c. Redundancy
 - d. Obsolescence
-

62. Current information availability is critical for:
- a. Historical archives
 - b. On-line transaction systems like reservations
 - c. Stand-alone tax software
 - d. Backup tapes
-

63. Which is NOT listed under “Additional DBMS Functionality”?
- a. Active processing
 - b. Visualization of data
 - c. Compiling Python scripts
 - d. System maintenance
-

64. Applications interact with databases via:
- a. Only reports
 - b. Queries and transactions
 - c. Hardware signals
 - d. Email
-

65. Unauthorized access prevention is a function of:
- a. Web browsers
 - b. DBMS security measures
 - c. Text editors
 - d. Printers
-

66. The “mini-world” in a hospital DB would include:
- a. Patient records and treatments
 - b. The building’s electricity
 - c. Staff lunch menus
 - d. Internet service provider
-

67. Which user designs canned transactions for naïve users?

- a. Tool developer
 - b. System analyst
 - c. Stand-alone user
 - d. Social media user
-

68. Business analysts increasingly work with:

- a. Paper files
 - b. Big Data and real-time data
 - c. Only SQL syntax
 - d. Hardware circuits
-

69. A user maintaining a personal photo database is a:

- a. Parametric user
 - b. Stand-alone user
 - c. DBA
 - d. System implementor
-

70. Which statement about NOSQL systems is noted in the text?

- a. They always require metadata
 - b. Some store data definition within structure, making it self-describing
 - c. They cannot handle multimedia
 - d. They replace all relational DBMS
-

True/False Questions (71–100)

71. A database is simply a collection of random files with no relationships.

- A. True
 - B. False
-

72. Data has implicit meaning and represents known facts.

- A. True
 - B. False
-

73. The DBMS software alone constitutes the entire “database system.”

- A. True
 - B. False
-

74. Multimedia databases are considered traditional applications.

- A. True
 - B. False
-

75. In the UNIVERSITY database, COURSEs are offered by DEPARTMENTs.

- A. True
 - B. False
-

76. Metadata is stored in the DBMS catalog.

- A. True
 - B. False
-

77. Program-data dependence means programs must change if data structure changes.

- A. True
 - B. False
-

78. Data abstraction hides storage details using a data model.

- A. True
 - B. False
-

79. Only one user can access a database at a time in a DBMS.

- A. True
 - B. False
-

80. OLTP systems support hundreds of concurrent transactions per second.

- A. True
- B. False

81. Database designers are “workers behind the scene.”

- A. True
 - B. False
-

82. Casual users access the database frequently for routine tasks.

- A. True
 - B. False
-

83. Stand-alone users often use ready-to-use packaged applications.

- A. True
 - B. False
-

84. System analysts design applications based on user requirements.

- A. True
 - B. False
-

85. Tool developers create software for database modeling and performance monitoring.

- A. True
 - B. False
-

86. Controlling redundancy is an advantage of the database approach.

- A. True
 - B. False
-

87. A DBMS cannot provide backup and recovery services.

- A. True
 - B. False
-

88. Enforcing integrity constraints is not a DBMS function.

- A. True

B. False

89. Using a DBMS always reduces application development time.

A. True

B. False

90. A DBMS is always necessary, even for simple single-user applications.

A. True

B. False

91. Embedded systems often cannot accommodate a general-purpose DBMS.

A. True

B. False

92. Real-time systems like telephone switches may avoid DBMS due to overhead.

A. True

B. False

93. DBMS can handle any data complexity, including genome databases, without limitation.

A. True

B. False

94. GIS applications may require special operations not supported by standard DBMS.

A. True

B. False

95. The database approach supports multiple views of the same data.

A. True

B. False

96. Social media users are considered end-users of database systems.

- A. True
 - B. False
-

97. Business analysts are classified as “actors on the scene.”

- A. True
 - B. False
-

98. The recovery subsystem ensures incomplete transactions are permanently recorded.

- A. True
 - B. False
-

99. NOSQL systems never use metadata.

- A. True
 - B. False
-

100. Economies of scale in DBMS come from consolidating data across departments.

- A. True
 - B. False
-

Answer Key

Multiple Choice Answers (1–70)

- 1. c
- 2. c
- 3. b
- 4. c
- 5. c
- 6. b
- 7. b
- 8. b
- 9. c
- 10. c

11. c

12. b

13. c

14. b

15. b

16. b

17. b

18. b

19. b

20. b

21. b

22. c

23. b

24. c

25. c

26. b

27. c

28. b

29. b

30. b

31. b

32. b

33. b

34. b

35. b

36. b

37. b

38. b

39. b

40. b

41. b

42. b

43. b

44. b

45. b

46. b

47. a

48. c

49. b

- 50. b
 - 51. c
 - 52. b
 - 53. a
 - 54. c
 - 55. b
 - 56. c
 - 57. b
 - 58. b
 - 59. b
 - 60. b
 - 61. b
 - 62. b
 - 63. c
 - 64. b
 - 65. b
 - 66. a
 - 67. b
 - 68. b
 - 69. b
 - 70. b
-

True/False Answers (71–100)

- 71. B (False)
- 72. A (True)
- 73. B (False) — includes data and possibly apps
- 74. B (False) — they are *recent*
- 75. A (True)
- 76. A (True)
- 77. B (False) — independence means *no* need to change
- 78. A (True)
- 79. B (False) — supports concurrency
- 80. A (True)
- 81. B (False) — they are “actors on the scene”
- 82. B (False) — casual = occasional
- 83. A (True)
- 84. A (True)

- 85. A (True)
 - 86. A (True)
 - 87. B (False) — it *does* provide them
 - 88. B (False) — it *is* a function
 - 89. A (True) — incremental time is reduced
 - 90. B (False) — not always necessary
 - 91. A (True)
 - 92. A (True)
 - 93. B (False) — modeling limitations may exist
 - 94. A (True)
 - 95. A (True)
 - 96. A (True)
 - 97. A (True) — as sophisticated/parametric end-users
 - 98. B (False) — only *completed* transactions are recorded
 - 99. B (False) — *some* NOSQL systems need no metadata, not all
 - 100. A (True)
-